

TALANK BARAL

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Overview

I am a PhD student in Computer Science at George Mason University specializing in software engineering. My research focuses on improving the speed and reliability of software testing and continuous integration using empirical methods and large language models (LLMs). I work on problems such as flaky test analysis, and test execution optimization, motivated by my prior industry experience with slow and brittle testing pipelines.

Publications

Talank Baral, Shanto Rahman, Bala Naren Chanumolu, Basak Balci, Tuna Tuncer, August Shi, and Wing Lam. “**Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation.**” 38th Annual International Conference on Automated Software Engineering, pages 901–913, (Kirchberg, Luxembourg), 2023. Acceptance rate: 21% (134/630) ASE’23 [1]

Talank Baral, Emirhan Oğul, Shanto Rahman, August Shi, Wing Lam. “**OptCD: Optimizing Continuous Development.**” 47th International Conference on Software Engineering, Demonstrations Track, pages 45–48, (Ontario, Canada), 2025. ICSE DEMO’25 [2]

Talank Baral, Abdelrahman Baz, Atish Kumar Dipongkor, August Shi, Kevin Moran and Wing Lam. “**Using Efficient Test Orders for Regression Testing.**” Under Review

Atish Kumar Dipongkor, **Talank Baral**, Laura Herrera-Perafan, Trevor Chandrapaul, Madhav Solanki, Carlos Lopez, Wing Lam, and Kevin Moran. “**Test Coverage Analysis of Agentic Pull Requests.**” Under Submission

Shanto Rahman, **Talank Baral**, August Shi, and Wing Lam. “**Reproducing Timing-Dependent Flaky Test Failures via a Template-Guided LLM Pipeline.**” Under Submission

Education

George Mason University 2022 – Now
PhD in Computer Science, CGPA: 3.98 (Anticipated Graduation: 05/27) Fairfax, VA, USA

Pokhara University 2015 – 2019
Bachelor’s in Software Engineering, CGPA: 3.67 Pokhara, Nepal

Experience

Graduate Research Assistant 2023 – 2025
George Mason University Fairfax, VA, USA

- Investigated CI and software testing inefficiencies using empirical analysis and LLM-based techniques, focusing on flaky test reproduction and repair, and test execution optimization

Graduate Teaching Assistant Fall 2022, Spring 2026
George Mason University Fairfax, VA, USA

- Worked as a GTA for the class CS262 (Introduction to Low-level Programming) during Fall 2022 and for class CS691 (Synergies of LLMs and SWE) during Spring 2026

Software Developer 2020 – 2022
JankariTech Pokhara, Nepal

- Helped different customers of JankariTech set up test automation
- Worked with OwnCloud, a client of JankariTech. OwnCloud is a German software company that works on content collaboration software. At work, my primary task included setting up test automation, CI pipelines, and writing and maintaining tests for different services of OwnCloud

Technical Skills

Research Areas: Software Testing, CI/CD Optimization, LLMs for Software Engineering

Programming Languages: Python, Bash, Java, C, C++, JavaScript, SQL

Test Automation: Selenium, Cypress, Nightwatch.js, Jest, Puppeteer, Behat

CI/CD & DevOps: GitHub Actions, GitLab CI, Travis CI, Drone, Docker

Grants and Awards

SIGSOFT CAPS Grant

Aug 2023

Awarded a travel reimbursement of \$1,200.00 for ASE'23. Didn't accept the offer because of visa issues

Summer 2023 Research Award - George Mason University

May 2023

Awarded by the College of Engineering & Computing, Department of Computer Science

First position in Project Expo at 7th GCES IT EXPO 2018

Feb 2018

Awarded by Gandaki College of Engineering and Science, Pokhara University, Pokhara

Undergraduate Academic Excellence Award

2015 - 2019

Merit-based Award provided by Gandaki College of Engineering and Science, Pokhara University, Pokhara

National Talent Identification Award (First Position Nationwide)

June 2012

Awarded by the Government of Nepal for designing a seed plantation machine

Notable Talks

ISSTA 2023 Doctoral Symposium (Doctoral Symposium)

Jul 2023

Optimizing Continuous Development and Predicting Test Information in Software Development

Seattle, WA, USA

Microsoft Redmond Campus (Industry Talk)

Jul 2023

Optimizing Continuous Development

Redmond, WA, USA

ESSEC 2024 (Colloquium Talk)

Jan 2024

Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation

William & Mary, VA, USA

ICSE 2025 (Conference Talk)

May 2025

OptCD: Optimizing Continuous Development

Ottawa, Ontario, Canada

Notable Open-Source Contributions

pact-cplusplus

Fixed compilation issues in pact-cplusplus in version 0.1.0 (<https://github.com/pact-foundation/pact-cplusplus>)

International Dataset of Flaky Tests (IDoFT)

Contributed to the IDoFT by adding and fixing some flaky tests (<https://github.com/TestingResearchIllinois/idoft>)

Several projects to optimize their CI builds

Submitted 25 PRs in several open-source projects, disabling generation of unnecessary directories in CI builds

OwnCloud

Automated several test cases for open-source Web and Desktop application of OwnCloud (<https://github.com/owncloud/client>)

avocode-email-tagsinput

Web-UI test automation for avocode-email-tagsinput (<https://github.com/avocode/avocode-email-tagsinput>)